VMware Migration

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May 11-14 2026

Red Hat

Migration

1. **VM metadata**
2. **Data copy**
   - curl (slow)
   - VDDK (medium)
   - Storage offload copy (fast)
3. **Guest conversion**
   - Mount guest OS
   - There will be always downtime

Demo

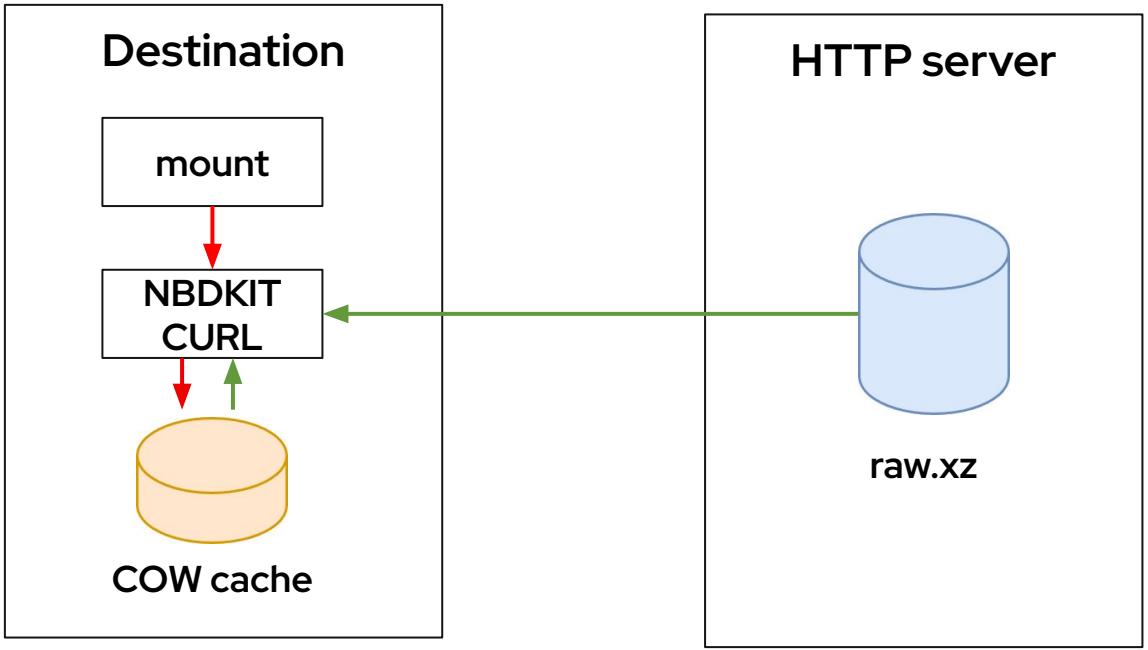
Why virt-v2v?

No changes on the source VM

- Removes VMware tools
- Installs virtio drivers
- Installs qemu guest agent
- Edits the guest to work with new drivers
- Rebuilds initramfs
- Preserve static IPs
- Preserve NIC name via udev rules (forklift)
- ...

Without VM credential

NBDKIT

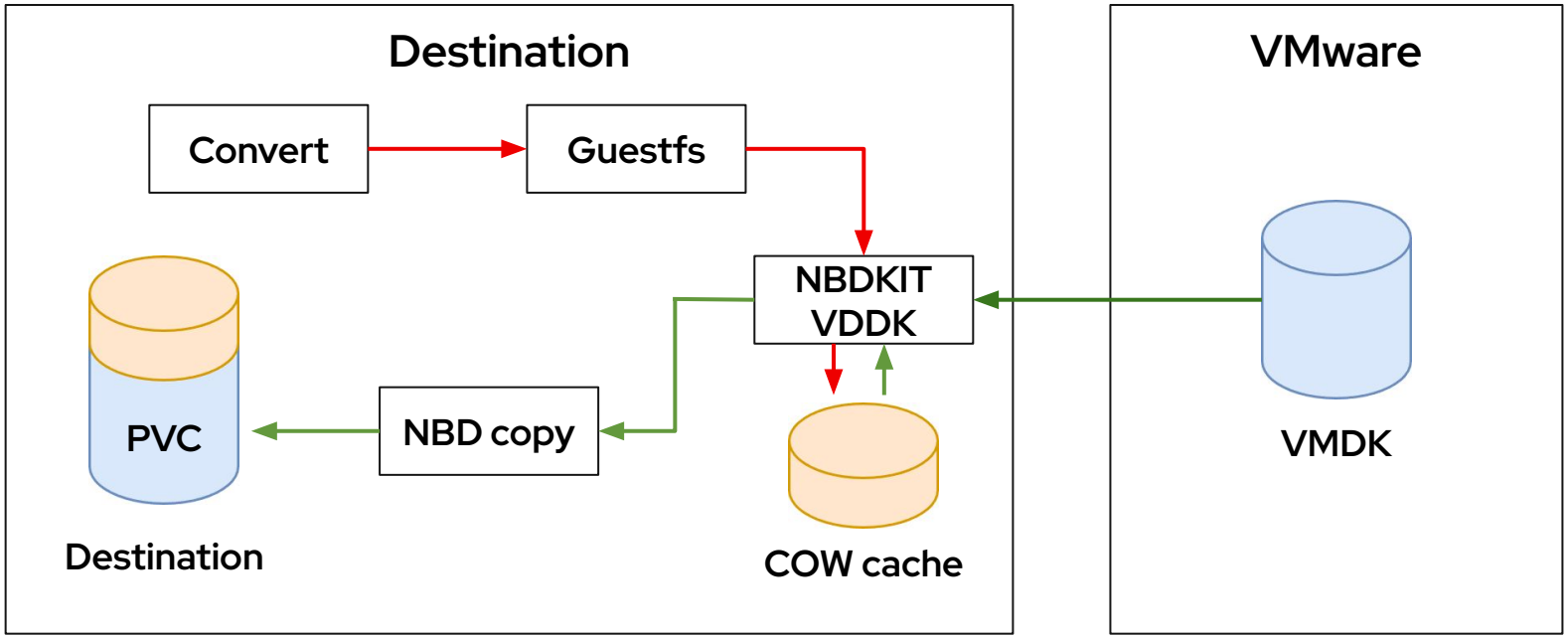


NBDKIT Demo

```
$ nbdkit
   -U /tmp/nbd.sock
   -P /tmp/nbd.pid
   --filter=cow
   --filter=xz
   curl http://localhost/Fedora-Workstation-Disk-44-1.7.aarch64.raw.xz

# sudo nbd-client -unix /tmp/nbd.sock /dev/nbd0
# sudo kpartx /dev/nbd0
# sudo mount /dev/nbd0p3 /mnt/fedora
# sudo vi /mnt/fedora/root/etc/fstab
```

How does virt-v2v work?



virt-v2v

Advantages:

- Fails fast
- Conversion supports many OS
  - Many linux distributions
  - Many windows versions
- Manages the disk transfer

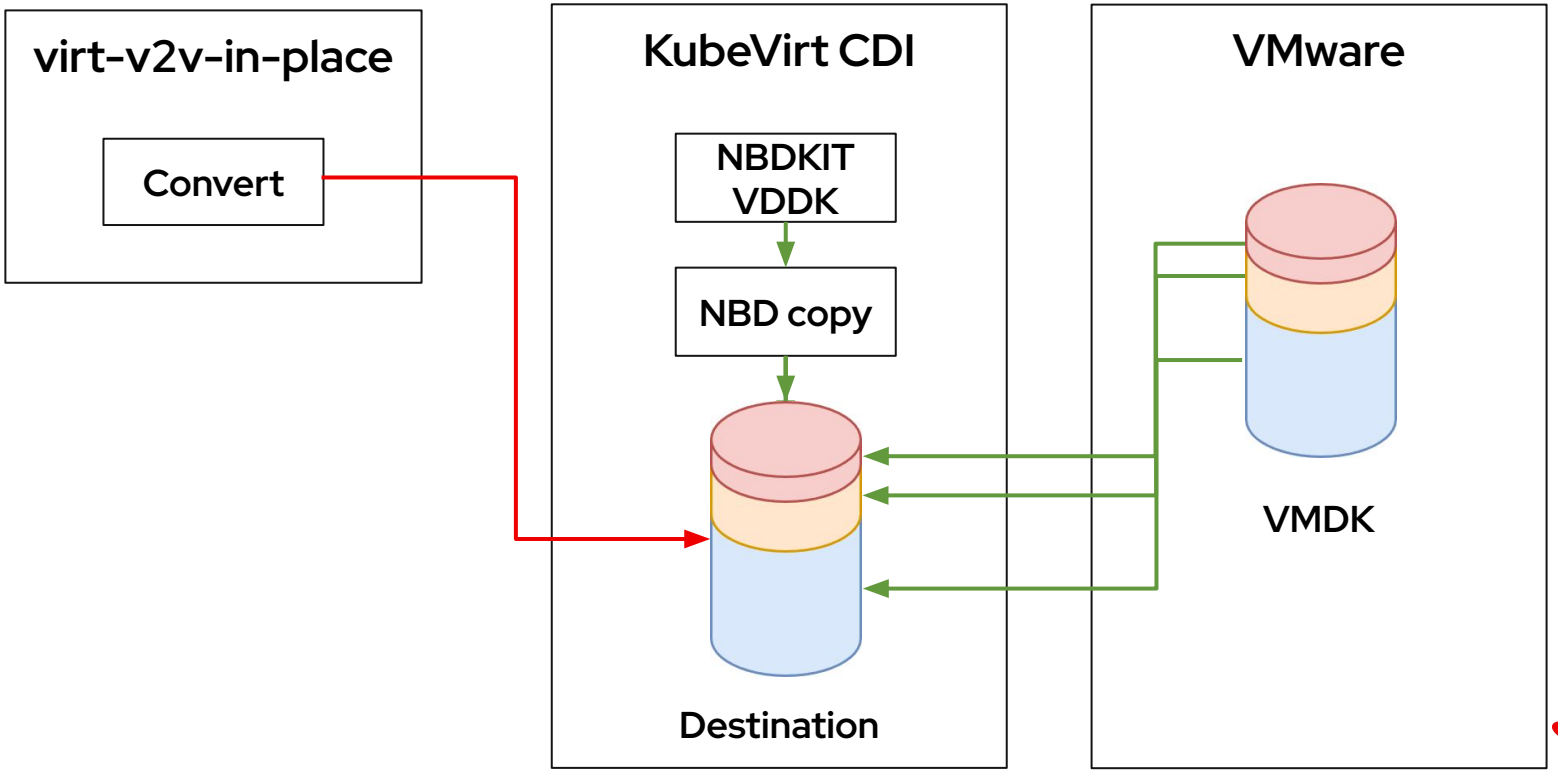
Disadvantages:

- High downtime, the VM needs to be turned off during whole migration
- Manages the disk transfer

Forklift

- User configure Kubernetes
  - Create storage classes
  - Create networks
- User adds a provider to Forklift
  - Forklift scans infrastructure
  - Validates VMs
- User creates a plan
  - Mappings (network/storage)
  - Migration type (cold/warm)
- Start migration

How does warm migration work?



Forklift warm migration

Advantages:

- Low downtime
- Removes the use of virt-v2v disk transfer
  - This allows us to transfer VMs which will fail conversion
  - Transfer shared disks
  - Offloading of the disk transfer

Disadvantages:

- High IO VMs won't converge
- Requires Change Block Tracking to be enabled on VM
- Migrations can bit take longer as there are many steps

VMware
Storage Offload

Red Hat

A month of overloaded network :(



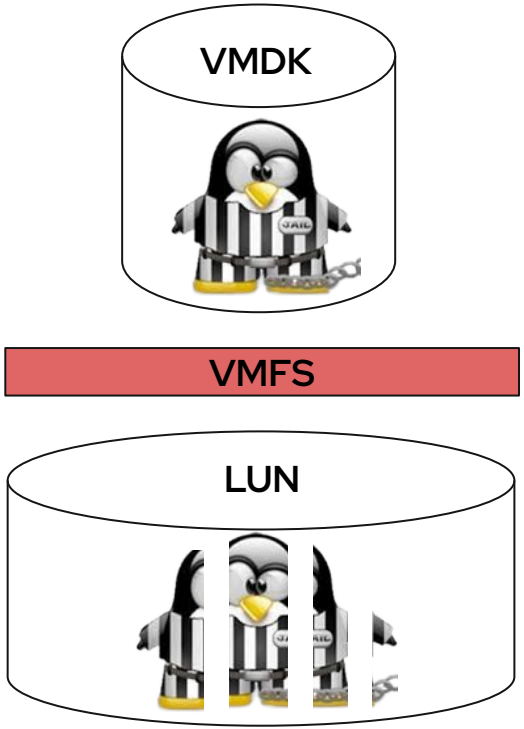
1 Terabyte of sad imprisoned vmdk over vmfs disk

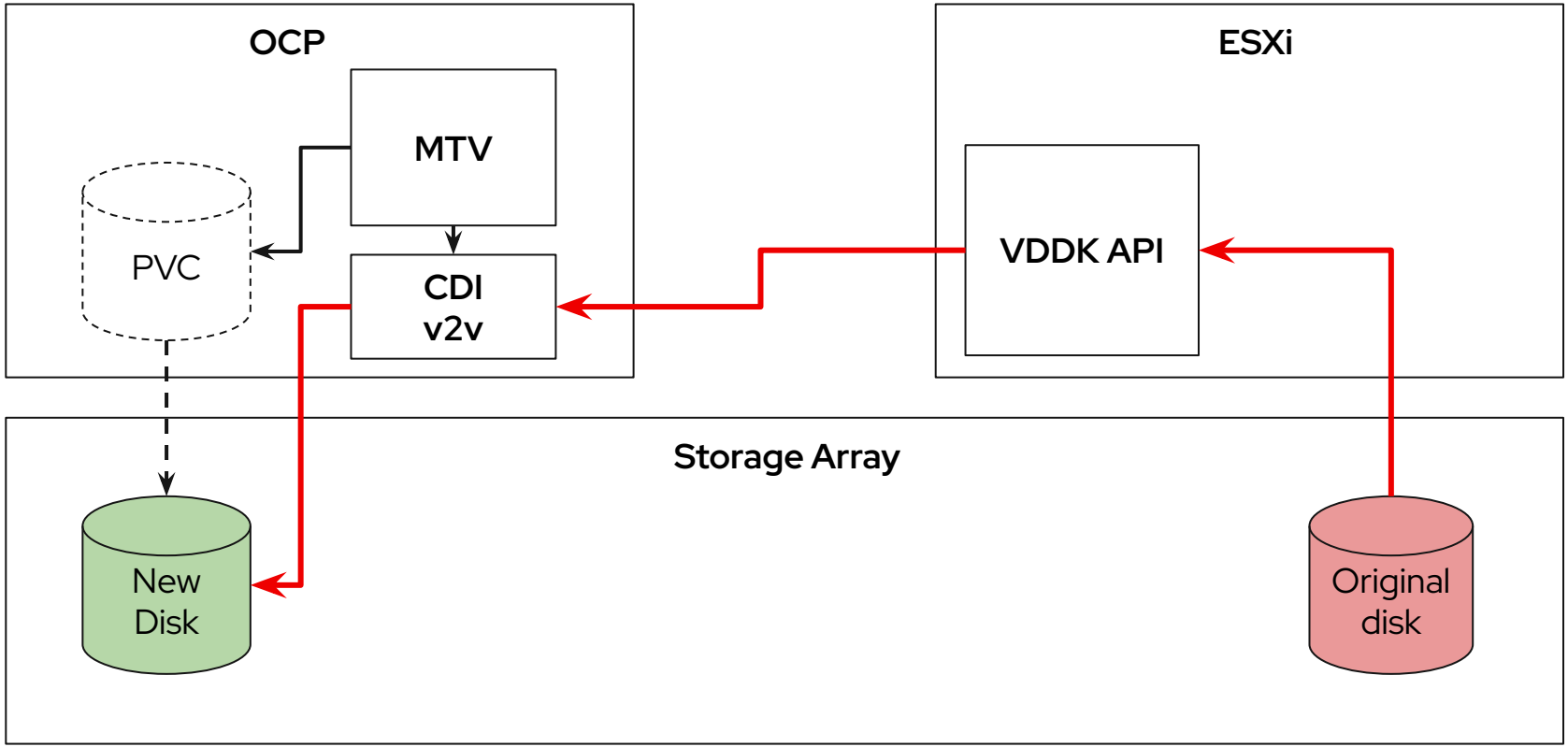


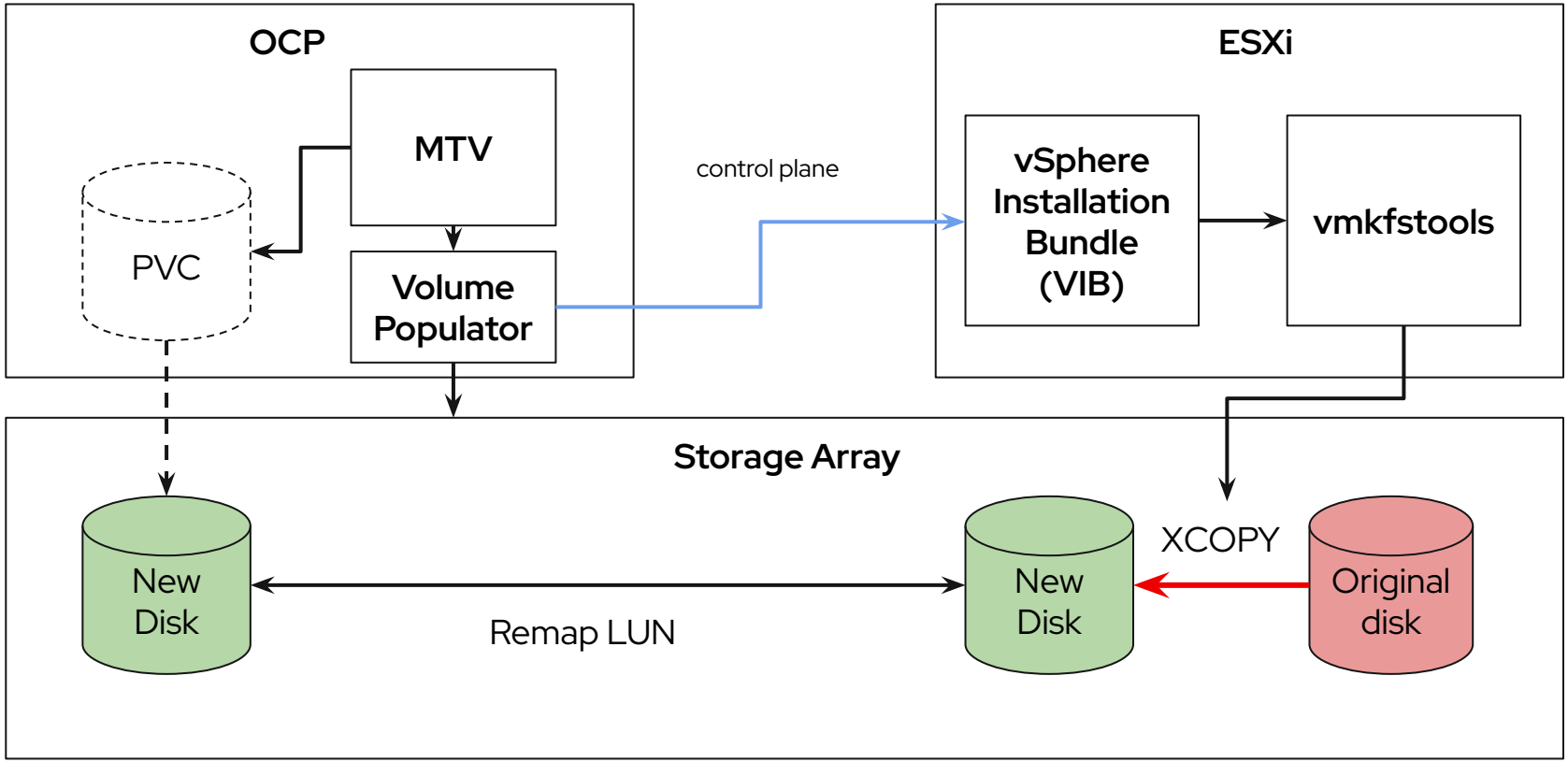


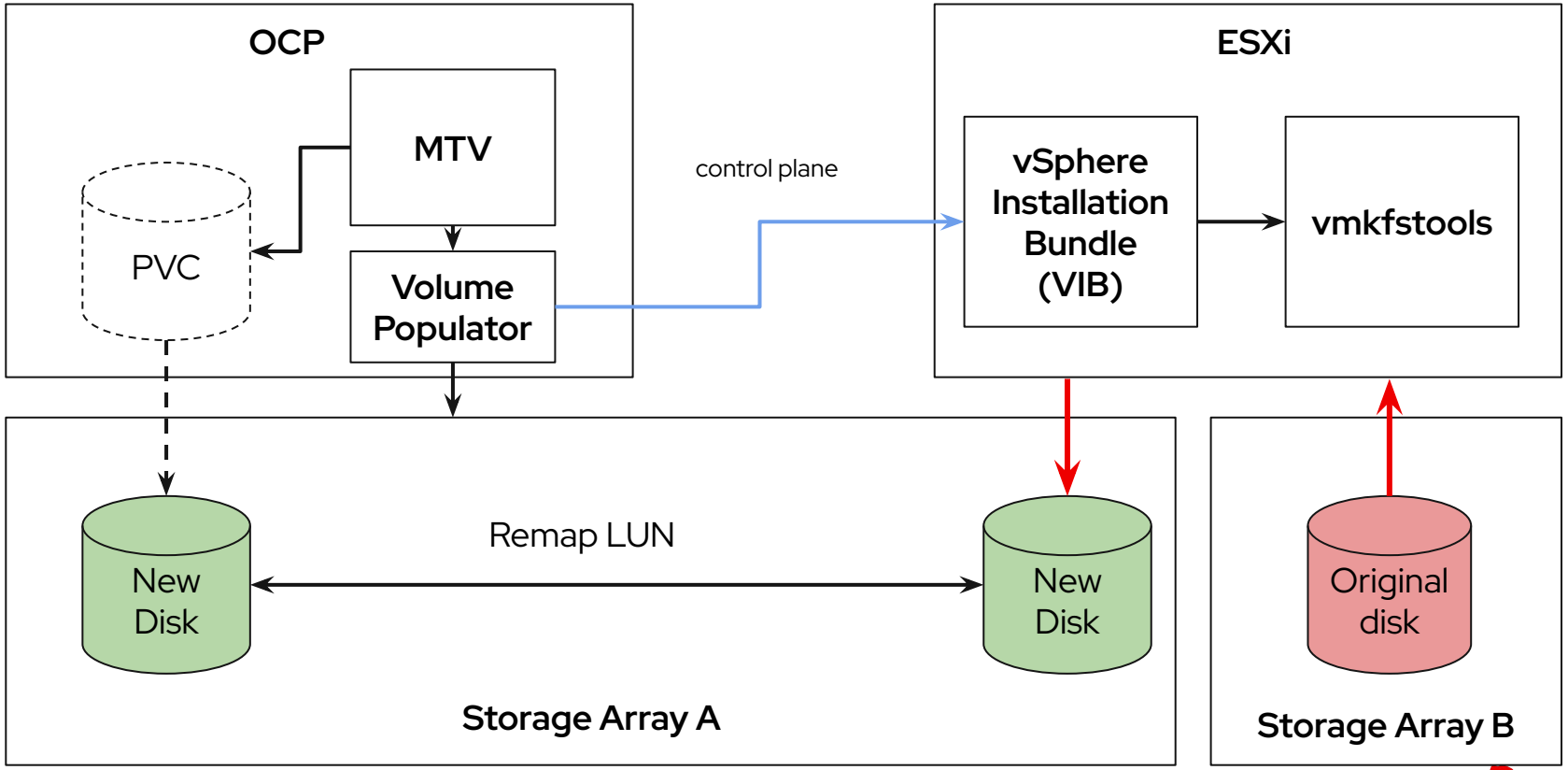
1 Terabyte of happy liberated disk

Fragmented penguin




OCP
MTV
PVC
CDI
v2v
ESXi
VDDK API
Storage Array
New
Disk
Original
disk


OCP
MTV
PVC
Volume Populator
control plane
ESXi
vSphere Installation Bundle (VIB)
vmkfstools
Storage Array
New Disk
Remap LUN
New Disk
XCOPY
Original disk


OCP
MTV
PVC
Volume Populator
control plane
ESXi
vSphere Installation Bundle (VIB)
vmkfstools
Remap LUN
New Disk
New Disk
Original disk
Storage Array A
Storage Array B

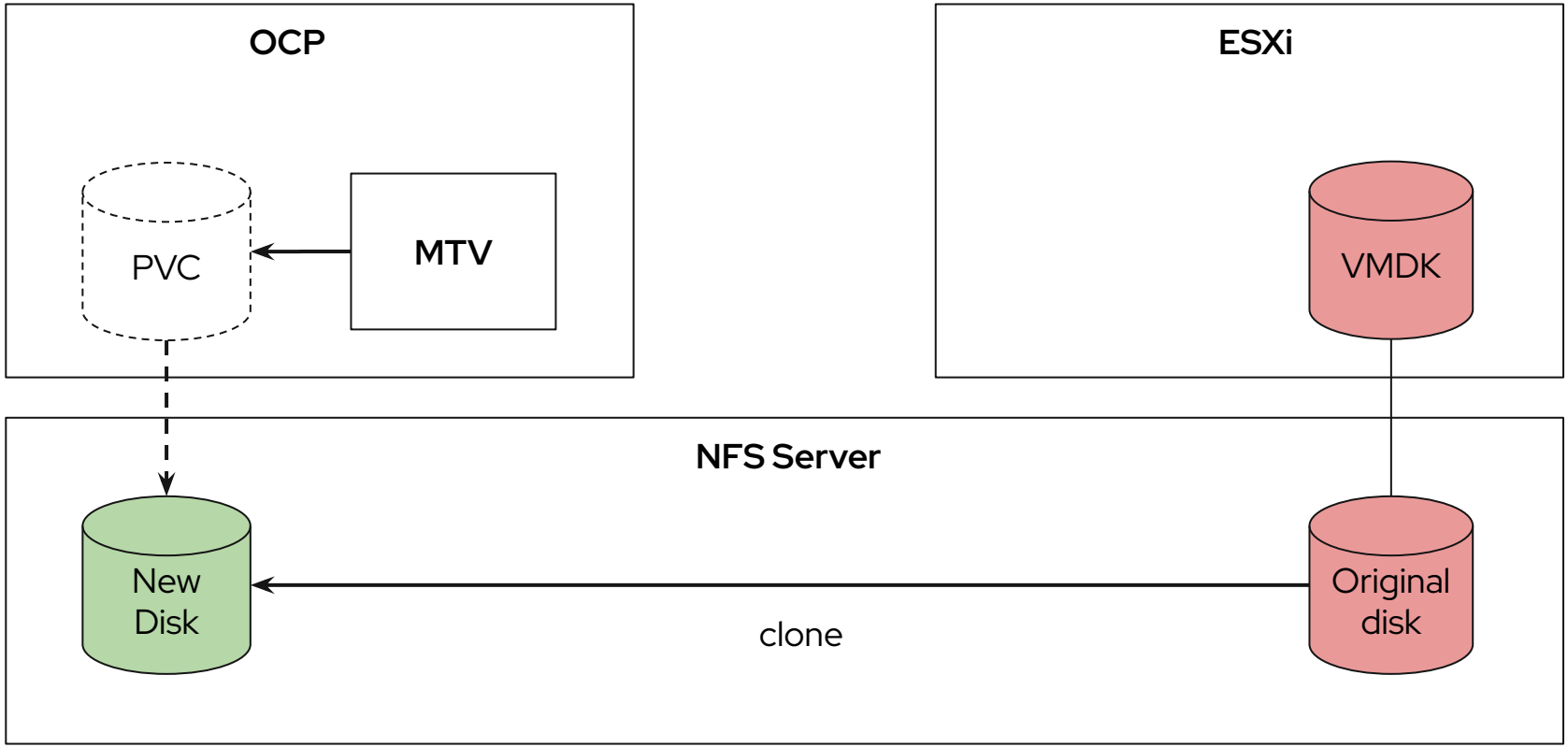
Summary of Copy Performance Results

Storage Array	500G	500G xcopy	x
Vantara	5700s	265s	x21.5
Ontap	4440s	780s	x5.7
3Par	4300s	490s	x8.7
Primera (copy fallback)	4200s	1000s	x4.2

- Performance results reflect variations in individual storage system characteristics.

Vendors

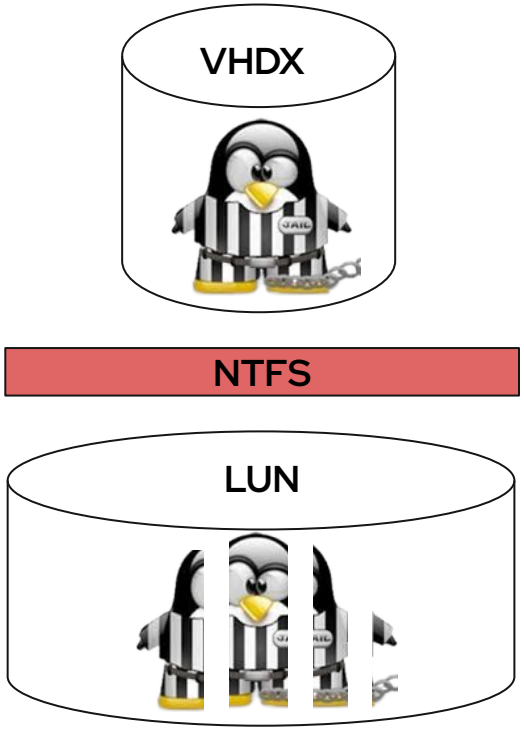
- Hitachi Vantara
- NetApp ONTAP
- HP Primera 3PAR
- Dell
  - Powerflex
  - Powerstore
  - Powermax
- Pure
- IBM FlashSystem

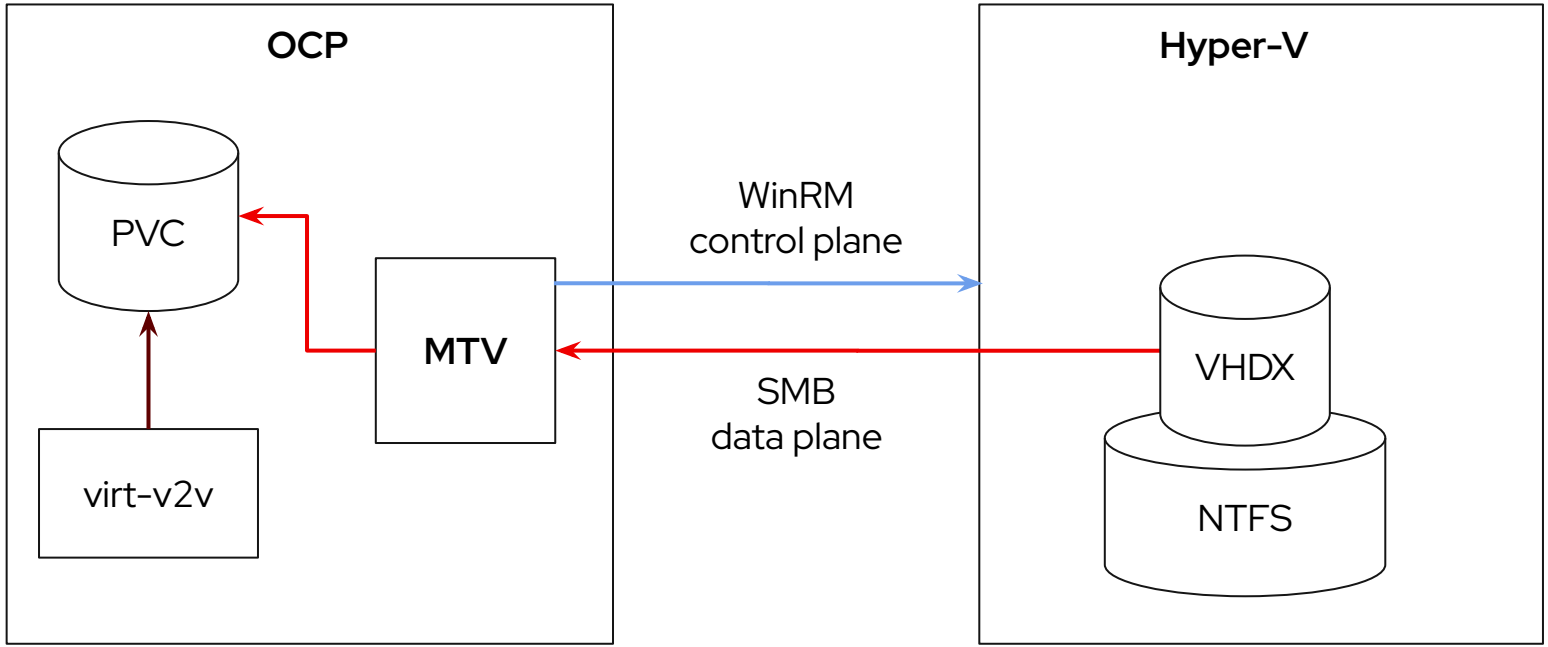

OCP
ESXi
PVC
MTV
VMDK
NFS Server
New Disk
clone
Original disk

Hyper-V Migration

Red Hat

Fragmented penguin




OCP
Hyper-V
PVC
MTV
virt-v2v
WinRM
control plane
SMB
data plane
VHDX
NTFS

GitHub



Survey

